Statistical Similarity Learning for Noise Image Pair Classification

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1 Introduction

This project addresses a binary classification problem in which pairs of noise images must be classified according to whether they originate from the same underlying distribution. Although noise images may appear visually similar, different stochastic generators often exhibit distinct statistical, spectral, and spatial properties. The task can therefore be interpreted as a learned two-sample test for stochastic signals.

Rather than treating the problem as a conventional image classification task, this work adopts a statistical perspective. Each noise image is modeled as a realization of a random field, and classification is performed by comparing distributional and structural descriptors extracted from the two images in each pair.

The objective of this project is to design a feature representation capable of capturing these properties and to evaluate multiple machine learning models for the noise matching task.

2 Dataset Description

The dataset consists of pairs of grayscale noise images stored as NumPy arrays. Each data sample contains two image identifiers and a binary label indicating whether both images originate from the same noise distribution.

The dataset is divided into three disjoint subsets:

- A training set used for model fitting.
- A validation set used for model selection and hyperparameter tuning.
- A test set used exclusively for final evaluation on Kaggle.

3 Feature Representation

The core challenge of the noise matching task lies in the fact that individual noise images are not deterministic objects but rather single realizations of underlying stochastic processes. As a result, direct pixel-wise comparison is unreliable, since two samples drawn from the same noise distribution are expected to differ substantially at the pixel level. To address this, the proposed approach represents each noise image using a set of handcrafted statistical descriptors that summarize its distributional, spectral, and spatial properties. Classification is then performed by comparing these descriptors across image pairs.

This feature representation can be interpreted as a finite-dimensional approximation of a statistical two-sample test, where the goal is to determine whether two empirical samples originate from the same probability distribution.

3.1 Feature Dimensionality Breakdown

Each noise image is represented by a fixed-length feature vector composed of multiple groups of handcrafted descriptors. The dimensionality of each feature group is explicitly controlled to balance expressiveness and model complexity. Table 1 summarizes the contribution of each component to the final representation.

Feature Group	Number of Features
Global statistical features	7
Histogram-based features	2
Frequency-domain (FFT) features	4
Patch-based spatial features	4
Total per-image features	**17**

Table 1: Per-image feature dimensionality breakdown.

For each image pair, per-image feature vectors are combined using multiple similarity and distance measures. Specifically, the absolute difference and squared difference are computed element-wise across all per-image features, contributing $2 \times 17 = 34$ dimensions. In addition, four scalar similarity measures are included: cosine similarity, Euclidean distance, Wasserstein distance, and Pearson correlation between raw pixel intensities.

As a result, each image pair is represented by a total of $34 + 4 = 38$ pairwise features, which form the input to the classification models.

3.2 Global Statistical Features

The first group of features captures global distributional properties of pixel intensities. Each image is flattened into a one-dimensional array, and the following statistics are computed: mean, standard deviation, variance, minimum, maximum, skewness, and kurtosis. These quantities correspond to the first four empirical moments of the underlying distribution, as well as its support.

These features characterize the overall shape of the noise distribution. While different noise generators may produce visually similar samples, they often differ in higher-order moments such as skewness or kurtosis, which describe asymmetry and tail behavior. Including these statistics allows the model to distinguish noise processes that share similar variance but differ in distributional shape.

3.3 Histogram-Based Distribution Features

To further describe the empirical distribution of pixel values, a normalized histogram is computed using a fixed number of bins. From this histogram, the entropy and variance are extracted. Entropy measures the degree of randomness or unpredictability in the noise, while histogram variance reflects how concentrated the probability mass is across intensity levels.

These features complement moment-based statistics by capturing global distributional characteristics that are not fully described by low-order moments. In particular, entropy provides a compact summary of the complexity of the noise distribution, which is often indicative of the underlying generator.

3.4 Frequency-Domain Features

Noise generators are frequently characterized by their spectral properties rather than their spatial appearance. To capture this information, frequency-domain features are extracted using

the magnitude of the two-dimensional discrete Fourier transform. The mean and variance of the Fourier magnitude spectrum are computed to summarize the overall energy distribution across frequencies.

Additionally, the spectrum is separated into low-frequency and high-frequency components. Low-frequency energy corresponds to large-scale or slowly varying structures, while high-frequency energy captures fine-grained or rapidly varying noise patterns. This separation allows the model to distinguish between noise processes that may share similar spatial statistics but differ in their spectral signatures.

3.5 Patch-Based Spatial Features

While global statistics describe the overall distribution of pixel values, they do not capture potential spatial non-uniformities. To model local variations, each image is divided into a fixed grid of non-overlapping patches. For each patch, the mean and variance of pixel intensities are computed.

The final patch-based features summarize the mean and variance of patch means and patch variances across the image. These descriptors provide information about spatial heterogeneity and local fluctuations in noise intensity. Such features are particularly useful for distinguishing between stationary noise processes and those exhibiting spatially varying behavior.

3.6 Pairwise Similarity Features

After extracting per-image feature vectors, classification is performed on image pairs by computing a set of pairwise similarity and distance measures. For each pair, the absolute and squared differences between corresponding per-image features are calculated. These differences quantify how strongly the statistical descriptors of the two images diverge.

In addition, cosine similarity is used to compare the orientation of feature vectors independently of their magnitude, while Euclidean distance provides a global measure of separation in feature space. To explicitly compare empirical distributions, the Wasserstein distance is included as a geometry-aware measure of distributional similarity.

Finally, the Pearson correlation coefficient is computed directly between the raw pixel values of the two images. This feature captures cross-correlation between noise realizations and provides a strong indicator of shared underlying structure.

3.7 Summary

The complete feature representation combines distributional statistics, spectral descriptors, spatial heterogeneity measures, and pairwise similarity metrics. By transforming each image pair into a fixed-length numerical representation that reflects statistical and structural similarity, the original problem is reduced to a standard binary classification task. This representation enables ensemble classifiers to learn effective decision boundaries without relying on pixel-level correspondence or deep architectures.

4 Machine Learning Models

To classify pairs of noise images based on their statistical similarity, two ensemble-based machine learning models were evaluated: Random Forest and XGBoost. Both models are well-suited for handling heterogeneous feature sets and non-linear decision boundaries, making them appropriate choices for the proposed feature representation. Using two distinct ensemble methods also allows for a comparative analysis of different learning paradigms.

4.1 Random Forest

Random Forest is an ensemble learning method based on the principle of bootstrap aggregation (bagging). It consists of a collection of decision trees, where each tree is trained on a bootstrap sample of the training data and considers a random subset of features at each split. Final predictions are obtained by aggregating the predictions of all trees, typically through majority voting for classification tasks.

This approach reduces variance and improves generalization, particularly when individual trees may overfit the training data. Random Forest is especially effective when the decision boundary can be expressed through a set of threshold-based rules, as is the case for many of the proposed similarity features. For example, strong indicators such as pixel-level correlation or distributional distance can be exploited through simple decision rules, which are then stabilized through ensemble averaging.

In this project, Random Forest serves as a robust baseline capable of leveraging strong, interpretable features without extensive hyperparameter tuning. Its inherent resistance to overfitting makes it well-suited for scenarios where a limited number of highly informative features dominate the classification task.

4.2 Random Forest Hyperparameter Tuning

To assess the sensitivity of the Random Forest classifier to its main hyperparameters, a series of controlled experiments was conducted on the validation set. The evaluated configurations varied the maximum tree depth, the minimum number of samples per leaf, and the number of trees in the ensemble, while keeping all other parameters fixed.

Figure 1 reports validation accuracy, Spearman correlation, and Kendall correlation across the tested configurations. Shallow trees (e.g., maximum depth of 2 or 4) result in noticeably lower performance, indicating underfitting due to insufficient model capacity. As tree depth increases, performance improves rapidly and stabilizes beyond a depth of approximately 16, suggesting diminishing returns from deeper trees.

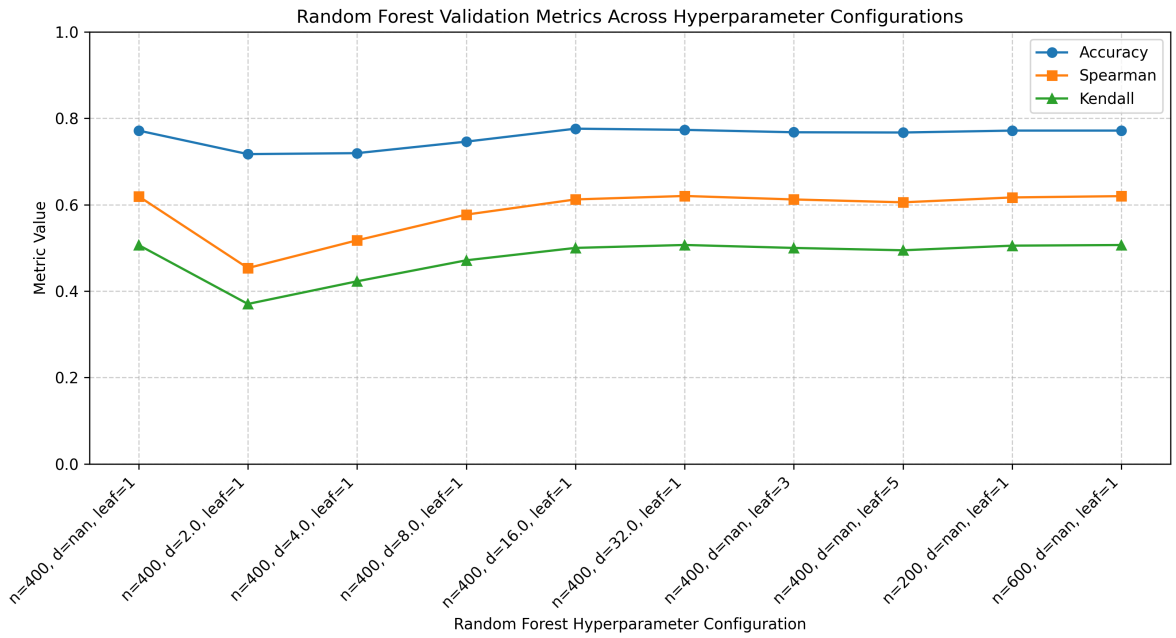

Figure 1: Validation accuracy, Spearman correlation, and Kendall correlation for Random Forest across different hyperparameter configurations.

Variations in the minimum number of samples per leaf and the number of estimators have a comparatively minor effect on performance once a sufficient ensemble size is reached. In particular, increasing the number of trees beyond 400 does not yield meaningful gains, while modest leaf regularization slightly smooths performance without improving accuracy.

Overall, the observed trends indicate that Random Forest performance is primarily driven by tree depth, while the model remains robust to reasonable changes in other hyperparameters. The final configuration was therefore selected based on a balance between performance and model simplicity.

4.3 XGBoost

XGBoost is an implementation of gradient boosting decision trees, where an ensemble of trees is constructed sequentially. Each new tree is trained to correct the errors made by the current ensemble by minimizing a differentiable loss function. In contrast to Random Forest, which trains trees independently, XGBoost builds trees in an additive manner, allowing it to model more complex feature interactions.

To control overfitting and improve generalization, XGBoost incorporates several regularization mechanisms, including shrinkage (learning rate), subsampling of training instances, and subsampling of features. These mechanisms encourage the model to learn gradually and prevent individual trees from dominating the final prediction.

In the context of this project, XGBoost is used as a complementary model to Random Forest. While Random Forest excels when a small number of strong features are sufficient to separate classes, XGBoost is capable of combining multiple weaker signals through successive refinements. Evaluating both models provides insight into whether the classification task benefits more from stable threshold-based decisions or from additive correction of residual errors.

4.4 XGBoost Hyperparameter Tuning

Hyperparameter tuning for XGBoost focused on the parameters most strongly affecting model capacity and regularization, namely tree depth, learning rate and number of estimators. Rather than performing an exhaustive grid search, parameters were varied in a controlled manner to isolate their individual effects on validation performance.

Figure 2 shows validation accuracy alongside Spearman and Kendall correlations for the tested configurations. Across all settings, performance remains relatively stable, indicating that the chosen feature representation provides strong discriminatory power independent of small hyperparameter changes. Moderate tree depths combined with learning rates in the range of 0.05–0.1 yield the best overall trade-off between accuracy and ranking consistency.

Adjustments to the number of estimators produce marginal gains once a sufficient number of boosting rounds is reached. Similarly, changes in subsampling and feature subsampling ratios do not significantly improve performance, suggesting that the model is not strongly overfitting and that regularization effects are already well balanced. The final configuration was selected from the plateau region of the validation curves, favoring stable performance over marginal improvements.

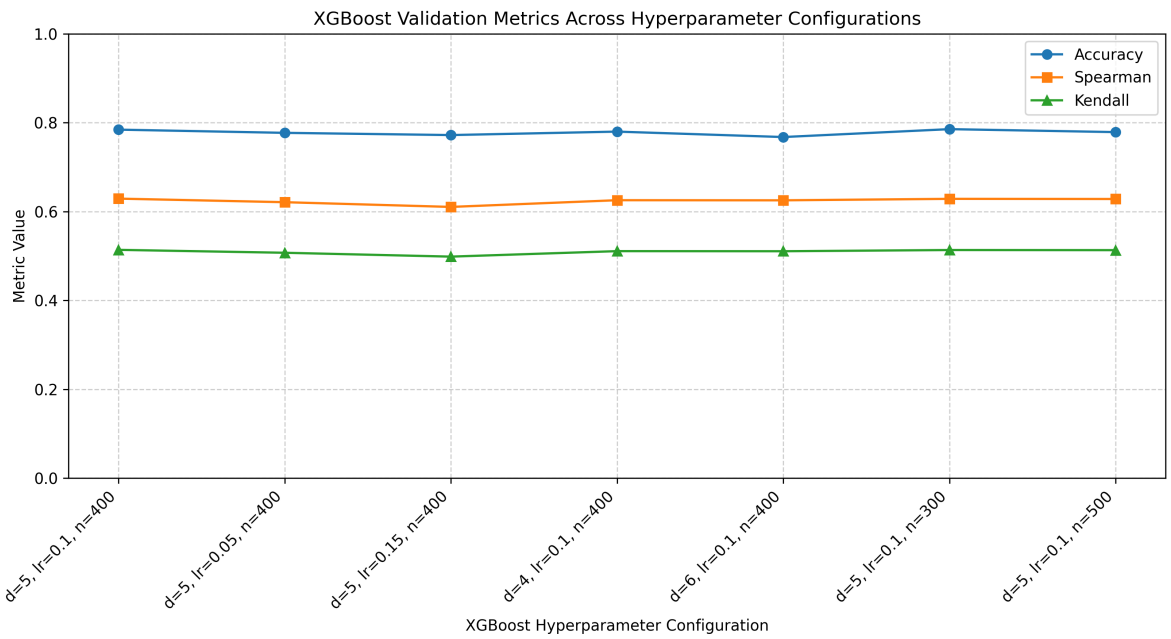

Figure 2: Validation accuracy, Spearman correlation, and Kendall correlation for XGBoost across different hyperparameter configurations.

4.5 Early Feature Representations and Unsuccessful Attempts

During the initial stages of this project, simpler feature representations were explored using the same ensemble-based machine learning models ultimately adopted in the final solution. These early attempts were motivated by the goal of establishing a baseline and understanding which aspects of the noise images were most informative for the matching task.

In both early variants, each noise image was normalized independently to zero mean and unit variance prior to feature extraction. Feature representations were limited to global statistical descriptors and frequency-domain summaries. Specifically, per-image features consisted of basic pixel statistics (mean, variance, skewness, kurtosis), histogram-based entropy, and coarse frequency-domain characteristics derived from the magnitude of the two-dimensional Fourier transform. For each image pair, absolute and squared differences between corresponding feature vectors were computed, along with cosine similarity and Euclidean distance.

Two closely related variants of this representation were evaluated. The first focused on a minimal implementation of the described feature set and direct training of XGBoost and Random Forest models. The second variant introduced a more structured processing pipeline and limited hyperparameter exploration, particularly with respect to the depth of the Random Forest ensemble. Despite these refinements, validation performance remained consistently low, with accuracies in the range of approximately 0.60–0.63 for both models.

The modest performance of these early approaches highlighted several limitations of the initial design. Per-image normalization was found to suppress absolute distributional information that is essential for distinguishing between different noise generators. Furthermore, the absence of spatial descriptors and direct signal-level similarity measures prevented the models from capturing important structural and correlation-based properties of the noise realizations. As a result, improvements in model capacity or hyperparameter tuning alone were insufficient to significantly enhance performance.

These observations motivated a redesign of the feature representation, leading to the removal of per-image normalization and the introduction of spatial patch statistics, distributional

distance measures, and pixel-level correlation features. The final feature set, described in Section 3, directly addresses the shortcomings identified in these early attempts and resulted in substantially improved validation and test performance.

5 Experimental Setup

All models were trained on the training set and evaluated on the validation set. Feature standardization was applied to ensure comparable feature scales. Performance was assessed using multiple metrics, including classification accuracy, mean absolute error (MAE), mean squared error (MSE), Spearman correlation, and Kendall correlation.

6 Results

Table 2 reports the performance of the evaluated models on the validation set.

Model	Accuracy	MAE	MSE	Spearman	Kendall
XGBoost	0.784	0.284	0.152	0.629	0.514
Random Forest	0.772	0.353	0.160	0.620	0.507

Table 2: Validation set performance metrics.

7 Discussion

Validation performance was consistently lower than the final Kaggle test performance. This discrepancy can be attributed to differences in difficulty between the validation and test sets. The validation set likely contains a higher proportion of ambiguous noise pairs, while the test set appears to be easier on average.

An additional factor influencing final performance is the retraining strategy applied after model selection. Once hyperparameters were fixed based on validation results, the selected models were retrained using the combined training and validation sets before generating final test predictions. This procedure increases the effective training set size and allows the model to better estimate feature distributions and decision boundaries. Retraining led to a noticeable improvement in test-set accuracy, particularly for the Random Forest model. This behavior is consistent with ensemble methods that benefit from larger and more diverse training data, as additional samples reduce variance and stabilize split decisions within individual trees.

This interpretation is supported by cross-validation experiments, which estimated performance around 0.75–0.77, and by the consistency between public and private leaderboard scores. The results indicate that the model generalizes well and that the observed performance gap is due to data distribution differences rather than overfitting.

8 Conclusion

This project demonstrated that noise image pair classification can be effectively addressed using handcrafted statistical and similarity-based features combined with ensemble learning methods. Random Forest achieved the highest performance, suggesting that the task is largely governed by strong threshold-based similarity cues.